Case Study - IV

Shri Sant Gajanan Maharaj College of Engineering, Shegaon
Department of Electrical (Electronics & Power) Engineering
Course Title & Course Code: High Voltage Engineering (8EP03)
Class: Final Year Electrical Engineering Semester: VIII
Name of the Course Teacher: Dr. R. S. Kankale
Title of the Innovative Practice: Use of YouTube video lectures for Blended and Flexible Learning
Activity Link: https://www.youtube.com/@ravikankale5583

Objectives/Goals of the practice:

The primary objective of this innovative teaching practice is to enhance accessibility, comprehension, and student engagement in complex topics of High Voltage Engineering by utilizing a dedicated YouTube channel. This method aims to:

- Enable self-paced learning by allowing students to access content anytime, anywhere.
- Reinforce classroom instruction through flipped learning and visual content delivery.
- Improve conceptual understanding with animations, diagrams, and real-life engineering examples.
- Provide revision-friendly material before exams and assessments.
- Extend the learning community beyond the classroom to reach broader audiences.

Use of Appropriate Methods:

To achieve the above objectives, the following methods were implemented:

- Creation of Subject-Specific Video Lectures: Videos were created on important topics such as breakdown theories of gaseous, liquid and solid dielectrics, overvoltage's. insulation coordination, corona, generation and measurement of different high voltage supply and testing of HV apparatus.
- Structured Playlists: Videos are organized in topic-wise playlists for sequential learning.
- Interactive Elements: Comments are encouraged for doubts and feedback, which are addressed by the faculty.
- In-Class Integration: Classroom sessions are used for deeper discussions and numerical problem-solving based on pre-watched video content.

Significance of Results:

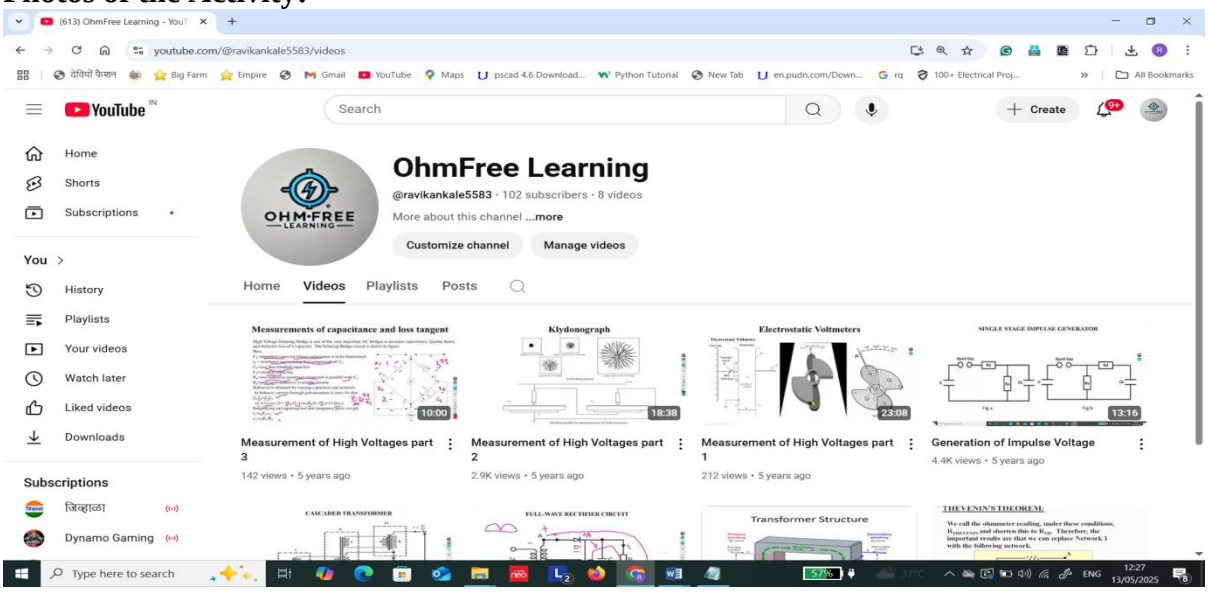
- Higher Accessibility: Students could revisit difficult topics and clarify doubts at their convenience.
- Improved Preparation: Videos served as quick revision tools before exams and tests.
- Wider Reach: Beyond institutional students, learners from other colleges also benefitted from the content.
- Enhanced Engagement: Students reported better focus and grasp of visually explained topics compared to traditional blackboard teaching.

Effective Presentation:

- Videos were presented using digital writing pads, animations, and clear voice explanations.

- Thumbnails and titles were crafted to make topic identification easier.
- Video analytics (views, likes, comments) were periodically reviewed to improve content delivery.
- Faculty introduced each topic in class and shared the video links in the student group.

Photos of the Activity:



POs Mapped: PO1, PO2, PO4, PO12

Reflective Critique:

Faculty observed increased student motivation and improved performance in internal assessments due to repeated access to video content. Students expressed appreciation for the ability to pause and rewatch explanations.

Peer Suggestions:

- Include quizzes within videos.
- Add subtitles for better accessibility.

Student Feedback:

- Videos were extremely helpful during exam time.
- Visual explanations made difficult concepts easier to understand.
- Concepts were taught in simple words which made learning very comfortable.

Evidences of success:

- Over 80% of students accessed the shared YouTube playlists.
- A noticeable improvement in internal test scores was observed for topics covered via video.
- Students requested similar video resources for other subjects.

Challenges faced during implementation:

- Some students faced internet connectivity issues in rural areas.

- A few students preferred classroom interaction and found video-only learning insufficient.
- Maintaining consistent video quality and clarity was initially challenging.
- Time investment in planning, recording, and editing was substantial.